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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 15.4

$$\begin{aligned}
 & \int x^2 e^{4x} dx \\
 & \left\{ \begin{array}{l} u = x^2 \\ dv = e^{4x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x dx \\ v = \frac{e^{4x}}{4} \end{array} \right. \\
 & = \frac{x^2 e^{4x}}{4} - \frac{1}{2} \int \frac{e^{4x}}{4} \cdot 2x dx \\
 & \left\{ \begin{array}{l} t = x \\ dw = e^{4x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} dt = dx \\ w = \frac{e^{4x}}{4} \end{array} \right. \\
 & = \frac{x^2 e^{4x}}{4} - \frac{1}{2} \left(\frac{x e^{4x}}{4} - \frac{1}{4} \int \frac{e^{4x}}{4} dx \right) \\
 & = \frac{x^2 e^{4x}}{4} - \frac{x e^{4x}}{8} + \frac{1}{8} \left(\frac{e^{4x}}{4} \right) + c \\
 & = \frac{x^2 e^{4x}}{4} - \frac{x e^{4x}}{8} + \frac{e^{4x}}{32} + c
 \end{aligned}$$

References